

APOLLO SOLAR VENTURES INC.

NUTECH Hardware & Software Solutions

NUTECH School System

gate.html

Two-Factor Attendance Station

Application Documentation

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Date: 22 June 2026

Contents

1. Overview

gate.html is the **entrance station**. A person taps their RFID card; the app identifies them, performs a **1:1 face match** against the face stored on their ACCOUNT (browser camera + face-api.js), then writes the attendance record — Time In/Out, Present or Tardy. It denies entry on a blocked/lost card or a face mismatch.

2. Flow

1. **Tap** → `gate-verify` returns identity, status, the face on file, and today's punches.
2. **Face match** → the browser compares the live camera face to the stored descriptor (euclidean distance $\leq$ threshold).
3. **Punch** → `gate-punch` writes the DTR row + an ATTENDANCE row.

3. Configuration

```
DEMO_MODE: true
N8N_BASE: ".../webhook/cafeteria"
CUTOFF: "08:00"           // tap after = Tardy
DEVICE: "GATE-01"
USE_CAMERA: false         // real face capture vs simulated
FORCE_FAIL: false         // demo the deny path
RFID_SRC: "hid"           // or "arduino"
ARDUINO_BAUD: 115200
MATCH_THRESHOLD: 0.5      // lower = stricter
```

4. Settings panel (⚙️)

- Demo mode toggle; Use real camera; Force face mismatch (deny demo).
- Tardy cutoff time; device name; webhook URL; Test connection.
- RFID source: USB HID or Arduino (Connect button + auto-reconnect).

5. Backend actions used

Action	Does
<code>gate-verify</code>	identity + face on file + today's punches
<code>gate-punch</code>	writes DTR (Time In/Out, Present/Tardy) + ATTENDANCE

6. Hardware

- **RFID:** USB HID or Arduino + PN532 over Web Serial (auto-reconnect, 115200).
- **Camera:** getUserMedia + face-api.js. Needs localhost/HTTPS, Chrome/Edge.

7. Running & go-live

4. **Serve over localhost/HTTPS** (Live Server or http.server).
5.  → uncheck **Demo mode** → Test connection.
6. Turn **Use real camera** ON only after faces are enrolled in the portal.

8. Notes

- **No face on file = no match.** Enrol faces via portal Register first; keep Use real camera OFF until then.
- **One reader per page** — the gate PC needs its own RFID reader.
- **Cutoff drives Present/Tardy** — a tap after CUTOFF is marked Tardy.